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SAAZA Unit 5: SAAZA Completion

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Purpose

- The purpose of this presentation is to guide prospective MMTIs on the preparation and submission of final survey products for review, approval, and database inclusion.



Learning Objectives

- Identify the required items to complete the AF IMT 3822, Landing Zone Survey Form
- Identify the required items to complete the AF IMT 4303, Helicopter Landing Zone Form
- Describe submission procedures for MAJCOM approval of SAAZA forms
- Describe submission procedures for inclusion in SAAZA databases



References

- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- MMT TACSOP



References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures



Outline

- Introduction / Overview
- Unit One: Mathematics and Measurement
- Unit Two: Landing Zone Criteria
- Unit Three: Soil Identification and Evaluation
- Unit Four: SAAZA Process
- Unit Five: SAAZA Completion



Outline

- Overview
- Landing Zone Survey Form
- Helicopter Landing Zone Form
- SAAZA Databases
- Review
- Summary



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AF IMT 3822

- Walkthrough

LANDING ZONE SURVEY	1A. LZ NAME Hawkeye LZ	1B. ZAR INDEX NO. UMCUSALZ00003-U	2A. COUNTRY United States of America	2B. STATE Arizona
	3. MAP SERIES/SHEET NUMBER/EDITION/DATE OF MAP WTI OPAREA SPECIAL MAP / Edition 2 / 9/2013			

- 1A: LZ Name (Bird, or Location)
- 1B: HQ AMC / A3DT will update after submission, if approved
- 2A: Country where LZ is located
- 2B: State, province or territory
- 3: Map series, sheet number, edition, date of map used (if used)



AF IMT 3822

- Walkthrough

4. SURVEY APPROVAL/DISAPPROVAL DATA			
4A. DATE SURVEYED 06 Oct 2020	TYPED NAME AND GRADE OF SURVEYOR James R. Carfora, CWO2	PHONE NUMBER (DSN) 269-2755	UNIT MAWTS-1, MCAS Yuma
4B. DATE REVIEWED 13 Oct 2020	TYPED NAME AND GRADE OF REVIEWER Charles Kuhnmuensch, Major	PHONE NUMBER (DSN) 2693539	SIGNATURE Charles Kuhnmuensch Digitally signed by Charles Kuhnmuensch Date: 2020-10-13 13:48:02 -07:00
4C. DATE 15 Oct 2020	TYPED NAME AND GRADE OF APPROVING AUTHORITY Col Steve Gillette, O-6	PHONE NUMBER (DSN) 269-2060	SIGNATURE Col Steve Gillette Digitally signed by Col Steve Gillette Date: 2020-10-15 20:37:56 -07:00
APPROVED ☒ DISAPPROVED ☐	UNIT AND LOCATION MAWTS 1, MCAS Yuma, AZ		

- 4A: Date physical survey was conducted, name, grade, phone number, and unit of Team Leader
- 4B: Date reviewed, reviewer name, grade, phone number, unit and location, with signature
 - User, Safety Officer, Squadron / ACE / MAGTF Commander
- 4C: Date approved / disapproved, name, grade, phone number, unit and location, with signature



AF IMT 3822

- Walkthrough

5. COORDINATING ACTIVITIES	
LZ CONTROLLING AGENCY OR UNIT MCAS Yuma AZ N/A	PHONE NUMBER (DSN) 928-269-3327
RANGE CONTROL Leg Iron, Yuma Range Control Facility, MCAS Yuma	PHONE NUMBER (DSN) 928-269-7080

- LZ Controlling Agency / Unit: This is YOU, if you are surveying for operations you will support
- Range Control: Info gained during Information Gathering, if agency exists; if not, may need to obtain a Land Use Agreement or Memorandum of Understanding – requesting unit responsibility



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- Walkthrough

6. LZ DIMENSIONS (FEET)			
LENGTH 3,511.67 ft / 1,070.35 m	WIDTH 60 ft / 18.28 m	APPROACH END OVERRUN LENGTH 300 ft / 91.44 m	DEPARTURE END OVERRUN LENGTH 300 ft / 91.44 m
LEFT CLEAR ZONE 35 ft / 10.66 m	LEFT SHOULDER 10 ft / 3.04 m	RIGHT CLEAR ZONE 35 ft / 10.66 m	RIGHT SHOULDER 10 ft / 3.04 m

- Unit of measure is feet
- Actual ground data, not the minimums required



AF IMT 3822

- Walkthrough

7. LZ AXIS DATA			
A. MAGNETIC 137.51° / 317.51°	B. GRID (UTM) 146.45° / 326.45°	C. TRUE 148.03° / 328.03°	D. SOURCE/DATE OF VARIATION DATA WMM 2020

- Primary run-in heading, use current year for date of variation if obtaining from GPS

8. GROUND POINT ELEVATION FOR RUNWAY	A. APPROACH END 541 ft MSL	B. DEPARTURE END 542 ft MSL	C. HIGHEST 542 ft MSL
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- MSL for each location



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- Walkthrough

9. LZ COORDINATES				
A. SPHEROID/DATUM WGS84 / WGS84		B. GPS DERIVED ☒ YES ☐ NO		C. GRID ZONE 11 S
		D. EASTING 07		E. NORTHING 36
F. LZ CENTER-POINT	MGRS COORDINATES 11S QS 74816 09044	WGS84 LATITUDE (D-M.MM) 32° 35.089' N	WGS84 LONGITUDE (D-M.MM) 114° 04.339' W	
G. APPROACH END	MGRS COORDINATES 11S QS 74520 09490	WGS84 LATITUDE (D-M.MM) 32° 35.335' N	WGS84 LONGITUDE (D-M.MM) 114° 04.520' W	
H. DEPARTURE END	MGRS COORDINATES 11S QS 75112 08598	WGS84 LATITUDE (D-M.MM) 32° 34.843' N	WGS84 LONGITUDE (D-M.MM) 114° 04.158' W	

- 9A/B: WGS-84 for GPS derived (almost always)
- 9C-E: Grid Zone, easting and northing obtained from map (different than MGRS)
- 9F-H: Grid zone, square designator, 10 digit grid, and lat / longs to the nearest 1/1000th minute



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- Walkthrough

10. LZ SURFACE DATA	
A. SURFACE Stabilized Soil	B. SOIL STRENGTH PROFILE Rwy 14/32 / Surface 7" CBR 35 / Subbase 10" CBR 24 / Subgrade CBR 26 (See Remarks)

- 10A: Type of surface (use letter identifier)
- 10B: Thickness and strength of each layer from the controlling point



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- Walkthrough

11. LZ LONGITUDINAL PROFILE	
A. GLIDE SLOPE RATIO Rwy 14: 55.4:1 / Rwy 32: 64.2:1	B. LONGITUDINAL RUNWAY GRADIENT Rwy 14: +0.03% / Rwy 32: -0.03%

- 11A: Glideslope ratio used to determine obstacle clearance for BOTH arrival and departure zones
- 11B: Percentage of slope for longitudinal gradient, use “+” for upslope and “-” for downslope



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- Walkthrough

12. TRANSVERSE SECTION GRADIENTS			
A. LEFT TRANSITION AREA +0.2%	B. LEFT GRADED AREA +0.2%	C. LEFT SHOULDER +0.1%	D. LEFT HALF RUNWAY +0.1%
E. RIGHT TRANSITION AREA +0.2%	F. RIGHT GRADED AREA +0.1%	G. RIGHT SHOULDER +0.1%	H. RIGHT HALF RUNWAY +0.1%
I. PENETRATIONS End of Usable (EOU) - Rwy 14 EOU: MGRS: 11S QS 74470 09567 / LL: 32° 35.377' N 114° 04.551' W / Elev: 541 ft MSL - Rwy 32 EOU: MGRS: 11S QS 75163 08521 / LL: 32° 34.801' N 114° 04.127' W / Elev: 542 ft MSL Controlling Obstacles (CO): *Distances referenced from Obstacle to Runway Threshold - Rwy 14 CO: 11 ft (AGL) Brush (11S QS 74401 09634 / Elev: 541 ft MSL) / 610 ft @ GSR 55.4:1 (1.03°) - Rwy 32 CO: 7 ft (AGL) Brush (Elev: 542 ft MSL) / 450 ft @ GSR 64.2:1 (0.89°)			

- 12A-H: Percent slope, “+” / “-” for up/downslope
- 12I: List any obstacles penetrations



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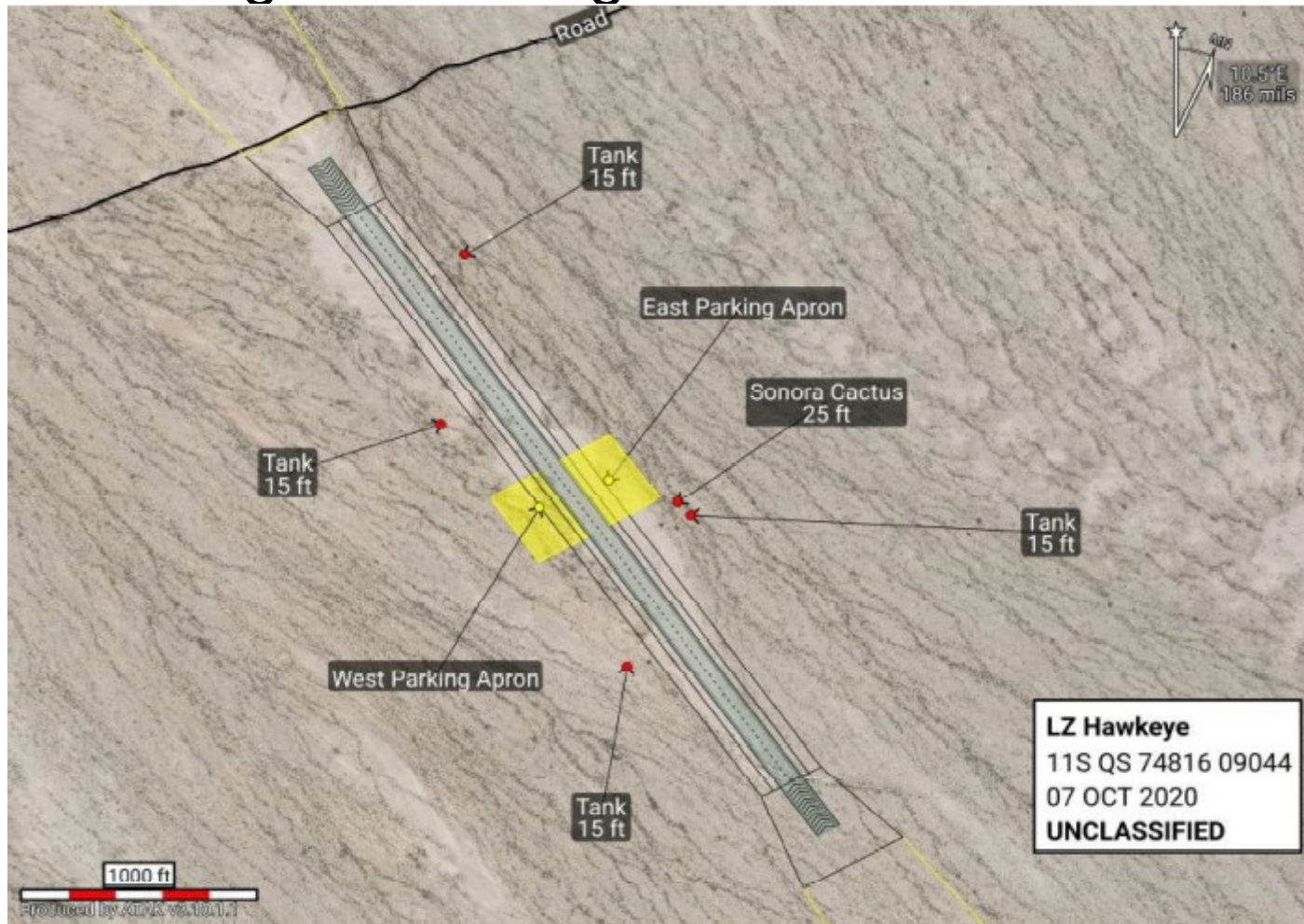
- Walkthrough – LZ Diagram





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- Walkthrough – LZ Diagram





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- Walkthrough - Remarks

Supported units commander accepts responsibility for damage to equipment, property, and/or injury to personnel.

Coordinates, elevation data derived from Trimble R8 GNSS.

- Horizontal margin of error +/- .5 meters, Vertical margin of error 1.5 X horizontal error.
- MAGVAR (10.51° E) Changing (0.09° West) per year derived from WMM 2020 as of 06 October 2020.

1. Administration / Coordination / Scheduling Instructions:

- a. LZ resides in the R-2301W range.
- b. For LZ use and scheduling contact Yuma Range Control.
- c. LZ was recently constructed in October 2020 with the use of Operational Dirt.

2. Obstacles/Hazards on the LZ: (All obstacles will be oriented, then referenced left or right of Rwy Centerline - All headings magnetic/see attached diagram) User or LZSO are responsible for identifying/ confirming ALL obstacles/hazards prior to conducting operations

- a. Rwy 14 / 32 overall SPACI Rating 95 = Green
- b. RCR: 20
- c. Dust / Green - Does not obstruct visibility
- d. Aggregate: None
- e. Rolling Resistance Material: None
- f. Jet Blast Erosion: None
- g. Stabilized Layer Failure: None
- h. Maintained Areas:
 - 1'-12' brush / +0' - 3500' / R 112'
 - 1'-12' brush / +0' - 3500' / L 102'

i. Clear Areas:

- Military Tank / +470' / L 278' / 15'
- Military Tank / +1091' / R 292' / 15'
- Sonora Cactus / +2048' / L 366'
- Military Tank / + 2155' / L 378' / 15'
- Military Tank / +2582' / R 277' / 15'

j. A - Zones: Un-restricted semi-improved road perpendicular to Rwy, 217' from EOU

k. B - Zones: None



AF IMT 3822

- Walkthrough - Remarks

3. Rapidly Rising/Mountainous Terrain (10 NM Radius):

- a. 690'MSL / Coyote Wash Hills / 1.05 NM / oriented South West
- b. 1192'MSL / Wellton Hills / 3.4 NM / oriented South West
- c. 3156'MSL / Gila Mountains / 8.25 NM / oriented South West
- d. 2888'MSL / Copper Mountains / 6.4 NM / oriented South East
- e. 1415'MSL / Baker Peaks / 4.9 NM / oriented North East
- f. 473'MSL / Radio Tower / 6.4 NM / oriented North West at 310°

4. Airspace (10 NM Radius):

- a. Hawkeye LZ is located in restricted area, specifically the R-2301W Ajo West, AZ - Boundaries. Beginning at the intersection of the United States-Mexican Border and long. 113°30'33"W.
- b. IR218 with westbound directional markings runs down to the boundary of the northern portion of R-2301W towards Yuma.
- c. V66 is a bi-directional airway north of the restricted area roughly 1.5KM and is transited by civil/military air traffic enroute to local destinations (Phoenix, Gila Bend).
- d. Northeast of the LZ has VR1267-1267A-1268 southwest bound directional routing.

5. Notes:

- a. There is no ARFF on site. Requests for ARFF support need to be coordinated with Yuma Range Control if needed.
- b. LZ is accessible via ground by 4x4 and tactical vehicles only.
- c. Road on 14 approach will require road guards during operations. This road is unrestricted and used for both government and recreational uses. Recreational users require permit issued by Yuma Range Control to operate within the range complex.



AF IMT 3822

- Walkthrough - Remarks

6. Rwy DCP Data processed with PCASE 2.09.06 / 22 DCP readings taken to a depth of 24 inches.
 - a. Controlling Reading CBR: 7" Surface CBR: 35 / 10" Subbase CBR: 24 / Subgrade CBR: 26
 - b. Controlling Layer: 10" Subbase CBR: 24
 - c. KC-130 / 175,000 lbs / 160,098 Passes
7. East Parking Apron (250' x 400') DCP Data processed with PCASE 2.09.06 / 1 DCP readings taken to a depth of 24 inches.
 - a. Controlling Reading CBR: 10" Surface CBR: 36 / 10" Subbase CBR: 44 / Subgrade CBR: 38
 - b. Controlling Layer: 10" Surface CBR: 35
 - c. KC-130 / 175,000 lbs / 293,531 Passes
8. West Parking Apron (250' x 400') DCP Data processed with PCASE 2.09.06 / 1 DCP readings taken to a depth of 24 inches.
 - a. Controlling Reading CBR: 9" Surface CBR: 26 / 10" Subbase CBR: 22 / Subgrade CBR: 32
 - b. Controlling Layer: 9" Surface CBR: 26
 - c. KC-130 / 175,000 lbs / 38,335 Passes



AF IMT 3822

- Walkthrough

15. PHOTOGRAPHY AVAILABLE

☒

YES

☐

NO

LOW LEVEL ROUTES

☒

NONE AVAILABLE

☐

ROUTE NAME/DESIGNATOR



Outline

- Overview
- Landing Zone Survey Form
- **Helicopter Landing Zone Form**
- SAAZA Databases
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AF IMT 4303

- Walkthrough

HELICOPTER LANDING ZONE SURVEY	1A. HLZ NAME LZ IVERSON	1B. ZAR INDEX NO. TBD	2A. COUNTRY WASABI	2B. STATE GINGER
	3. MAP SERIES/SHEET NUMBER/EDITION/DATE OF MAP WTI OPAREA SPECIAL MAP / EDITION 002 / 2017			

- 1A: LZ Name (Bird, or Location)
- 1B: HQ AMC / A3DT will update after submission, if approved
- 2A: Country where LZ is located
- 2B: State, province or territory
- 3: Map series, sheet number, edition, date of map used (if used)



AF IMT 4303

- Walkthrough

4. SURVEY APPROVAL/DISAPPROVAL DATA			
4A. DATE SURVEYED 20190918	TYPED NAME AND GRADE OF SURVEYOR Montoya, Inigo; O-2, USMC	PHONE NUMBER (DSN) 928-269-0604	UNIT VMM-999 / 99 th MEU
4B. DATE REVIEWED 20191026	TYPED NAME AND GRADE OF REVIEWER Vizzini, Wallace; O-3, USMC	PHONE NUMBER (DSN) 928-269-1527	SIGNATURE 
	UNIT AND LOCATION VMM-999, 99 th MEU, USS Revenge		
4C. DATE 20191031	TYPED NAME AND GRADE OF APPROVING AUTHORITY Roberts, Westley; O-6, USMC	PHONE NUMBER (DSN) 928-269-4146	SIGNATURE 
APPROVED	DISAPPROVED	UNIT AND LOCATION 99 th MEU, USS Revenge	

- 4A: Date physical survey was conducted, name, grade, phone number, and unit of MMT Leader
- 4B: Date reviewed, reviewer name, grade, phone number, unit and location, with signature
 - User, Safety Officer, Squadron / ACE / MAGTF Commander
- 4C: Date approved / disapproved, name, grade, phone number, unit and location, with signature

Squadron / ACE / MAGTF Commander



AF IMT 4303

- Walkthrough

5. COORDINATING ACTIVITIES	
HLZ CONTROLLING AGENCY OR UNIT 99 th MEU / VMM-999 / MACG-98 / MMT "FEZZIK"	PHONE NUMBER (DSN) 928-269-0604
RANGE CONTROL Wasabian National Range Agency – "Pepper" – VHF 111.111 / UHF 333.333	PHONE NUMBER (DSN) 928-269-5555

- LZ Controlling Agency / Unit: This is YOU, if you are surveying for operations you will support
- Range Control: Info gained during Information Gathering, if agency exists; if not, may need to obtain a Land Use Agreement or Memorandum of Understanding – requesting unit responsibility



AF IMT 4303

- Walkthrough

6. HLZ AXIS DATA (APPROACH/DEPARTURE)			
A. MAGNETIC 281°	B. GRID (<i>UTM</i>) 270°	C. TRUE 270°	SOURCE/DATE OF VARIATION DATA GPS / 2019

- Primary run-in heading, use current year for date of variation if obtaining from GPS

7. HLZ COORDINATES				
A. SPEROID/DATUM WGS84/NAD8	B. GPS DERIVED ✗ YES NO	C. GRID ZONE 11N	D. EASTING 726666	E. NORTHING 3615767
F. HLZ CENTER-POINT	MGRS COORDINATES 11S QS 26666 15767	WGS84 LATITUDE (<i>D-M.MM</i>) N 32:39.23		WGS84 LONGITUDE (<i>D-M.MM</i>) W 114:34.60

- 7A/B: WGS-84 for GPS derived (almost always)
- 7C-E: Grid Zone, easting and northing obtained from map (different than MGRS)
- 7F: Grid zone, square designator, 10 digit grid, and lat / longs to the nearest 1/1000th minute



AF IMT 4303

- Walkthrough

8. HLZ SURFACE DATA			
A. LENGTH (FEET) 325'	B. WIDTH (FEET) 420'	C. ELEVATION 210' MSL	D. QTY/TYPER (ex. 2/H53: 2/H60) 3 UH-1 or 2 MV-22
E. QUADRANT (OBSTRUCTED/UNOBSTRUCTED) N and S Quadrants Obstructed		F. SLOPE 14:1 Approach axis, 11:1 Departure axis	
9. REMARKS 1. N Quadrant obstructed by 45' palm trees from 310 radial to 035 radial 2. S Quadrant obstructed by 45' palm trees from 070 radial to 220 radial 3. LZ is grass covered, and poses no threat for brown-out			

- 8A/B: LZ Dimensions
- 8C: Enter in feet MSL
- 8D: Maximum T/M/S of requesting agency
- 8E: Obstructions to approach and departure
- 8F: LZ slope from approach angle
- 9: Describe items from 8E, and other concerns



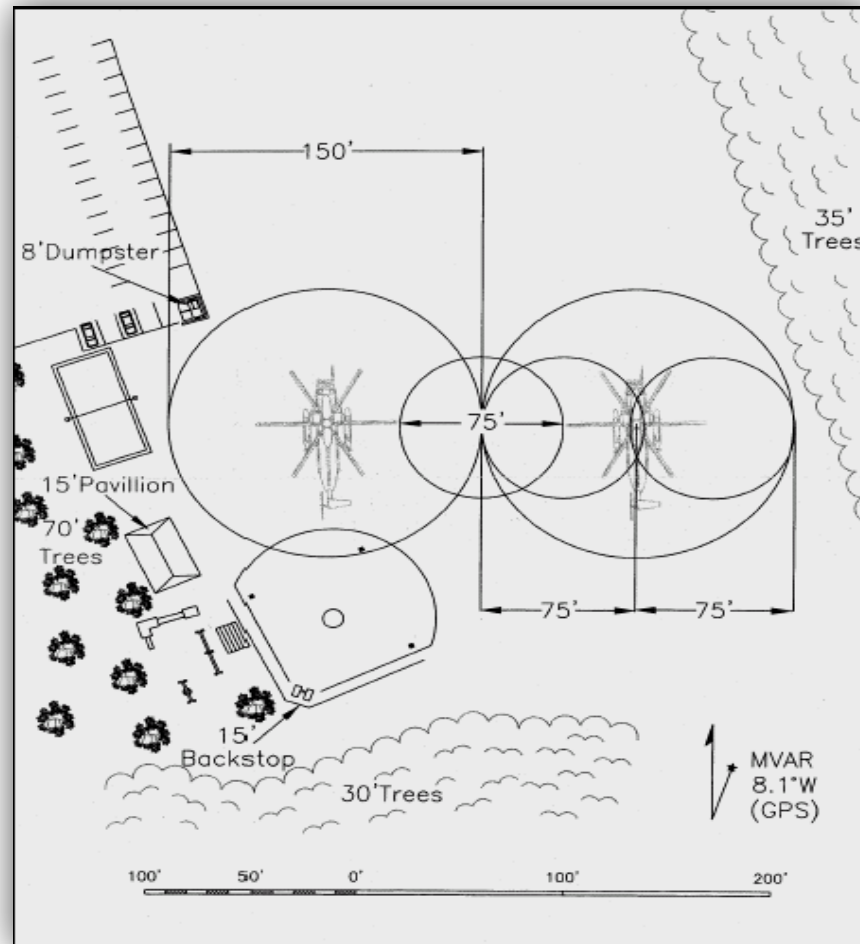
AF IMT 4303

- Walkthrough - Remarks



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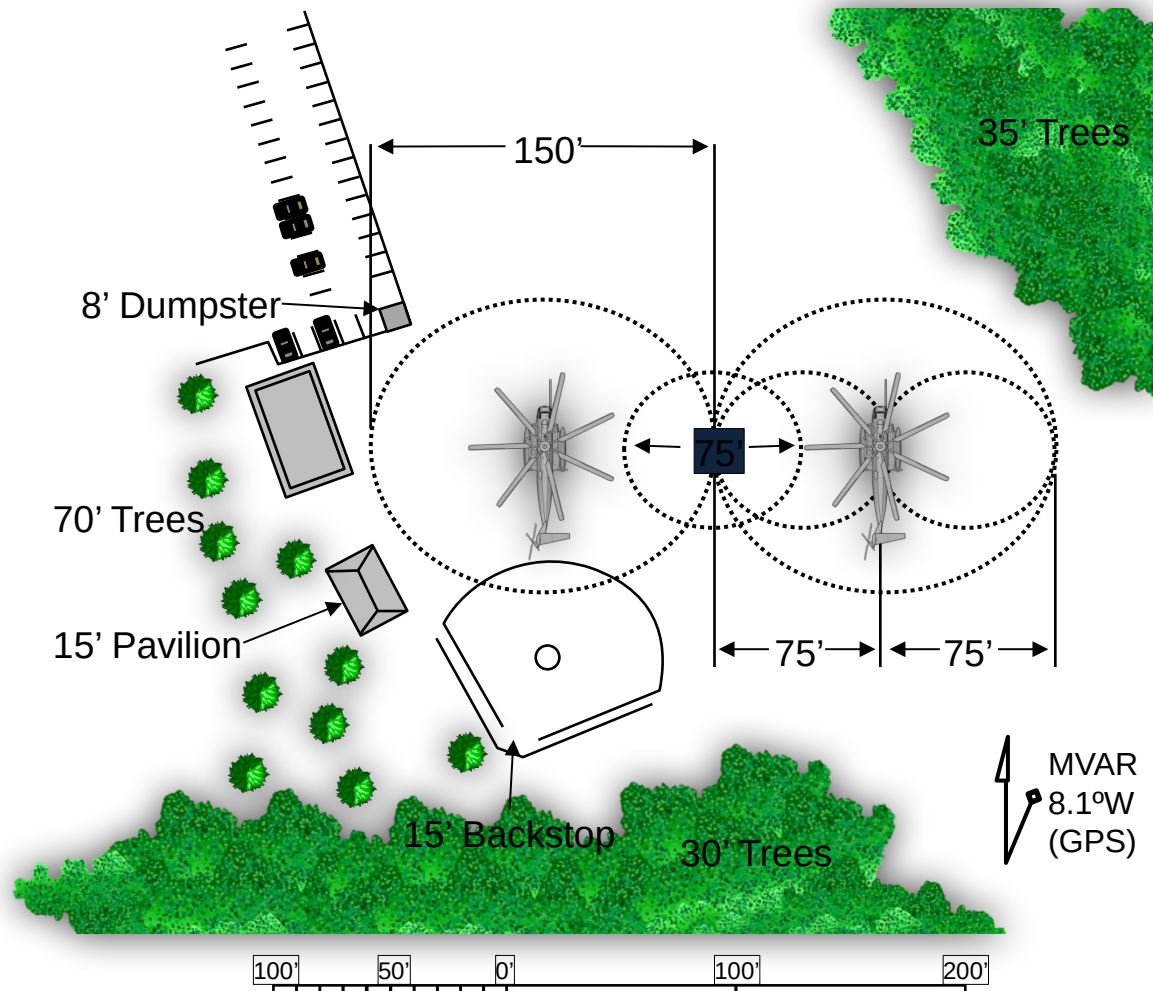
- Walkthrough – HLZ Diagram





AF IMT 4303

- Walkthrough – HLZ Diagram





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SAAZA Databases

- Talon Point

 TALON POINT
UNCLASS

JAMES CARFORA
USMC, MAWTS 1 

Welcome, James Carfora

 HOME

 SEARCH

 WORKFLOW

 MY ACCOUNT

 INFORMATION

 HELP

 LOG OUT

 5 NOTIFICATIONS

← HIDE PANEL

Find Surveys by Keyword

e.g. MAJCOM, Country, Type of Survey

Q

☒ Published ☐ In Progress

 DZ

 FARP

 HLZ

 LTFW

 LZ



Ready to create a new survey? [Add Survey](#)



Terms

Talon Point 2021 - v34.0.0-3



SAAZA Databases

- Talon Point Routing
 - Survey Completed by MSAAZA / SAAZA Marine
 - Routed for QC to EAF OIC / MAW Rep that has MSAAZA / SAAZA
 - QC Routed the survey to the MAG ASO / MAW DOS for Safety of Flight review
 - ASO routes the Survey to MAG CO / MAW CG
 - MAG CO / MAW CG approved / Disapproved the Survey



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Questions / Critiques



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